Predicting Campus Car Park Availability with Calibrated Uncertainty

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Code and data: github.com/taka8rie/ecen7102-carpark

Workload & Contribution Statement

Member	Main tasks	Share
[Name 1]	[e.g. data pipeline, cleaning and feature engineering (Steps 1–2); data quality analysis]	[%]
[Name 2]	[e.g. baselines, Ridge and XGBoost models (Steps 3–4); interval methods (Steps 5–6)]	[%]
[Name 3]	[e.g. literature review, evaluation and figures (Step 7); report writing and slides]	[%]

All members reviewed the code and the final report and can explain every part of it. Generative AI was used as described in Appendix A.

Abstract

Drivers arriving at the University of Macau (UM) cannot see in advance whether a staff space will be free. We forecast the number of available staff car spaces at car park P5 one hour and 24 hours ahead, at 15-minute resolution, and attach 80% prediction intervals. Using 2023 open data from the UM Data and API platform, we train on 1 January–15 September, select models on 16–30 September, calibrate intervals on October, and evaluate once on a walk-forward simulation of November–December.

Predicting the change from the current value, rather than the level, makes gradient-boosted trees robust to a drift of about 30 spaces in the series' level. XGBoost reduces daytime mean absolute error (MAE) from 11.2 to 6.2 spaces at 1 h and from 21.1 to 16.1 spaces at 24 h, relative to the best naive baseline. Multi-quantile XGBoost gives intervals that widen and narrow with the time of day, but they cover only 77% (1 h) and 61% (24 h) of test values. Split conformal calibration on October restores coverage to 80–86% (1 h)

and 73–80% (24 h). Hour-wise (Mondrian) conformalized quantile regression gives the best interval score at 1 h; a fixed-width conformal band is best at 24 h, where uncertainty about the day's overall level dominates. A one-sided conformal bound of the form "at least L spaces with 90% confidence" held for 90.6–93.1% of test points.

1 Introduction and Literature Review

1.1 Motivation

Campus parking is scarce at the hours when most staff arrive. On a teaching weekday, available staff spaces at UM car park P5 fall from about 130 at 07:00 to about 60 by 11:00. A driver who knows in advance whether spaces will be left can choose another car park, leave earlier or take the shuttle. A single predicted number is not enough for that decision: "about 70 spaces" is reassuring only if the forecast is unlikely to be far too high. This project therefore forecasts both the number of available spaces and a calibrated range around it.

1.2 Parking availability prediction

Parking prediction has been studied mainly with sensor or transaction data from city car parks. Vlahogianni et al. [1] built a real-time prediction system for smart cities that uses multilayer perceptron neural networks to predict the occupancy of on-street parking areas. Zheng, Rajasegarar and Leckie [2] compared regression trees, neural networks and support vector regression on car park data from San Francisco and Melbourne, and found the regression tree the most accurate of the three [22]. Camero et al. [3] used recurrent neural networks tuned by an evolutionary algorithm for car park occupancy. Yang et al. [4] combined graph convolutional and LSTM networks to predict block-level occupancy in Pittsburgh 30 minutes ahead, adding traffic speed and weather data. Zhao and Zhang [5] compared linear regression, support vector regression, neural networks and ARIMA across parking types and scales in three Chinese cities, and found support vector regression the most stable. Provoost et al. [6] compared a neural network with a random forest against a seasonal random walk for a garage in the Netherlands, using a chronological train, validation and test split as we do.

These studies report point forecasts and point-error metrics. Uncertainty is rarely quantified. A recent exception is Nezhadettehad et al. [7], who used Bayesian neural networks to separate epistemic and aleatoric uncertainty for Melbourne on-street parking. Their intervals depend on the model's distributional assumptions, and coverage on new data is not guaranteed.

1.3 Uncertainty quantification and conformal prediction

Quantile regression [11] estimates conditional quantiles directly by minimising the pinball loss, and gradient boosting [8, 9] can optimise this loss. The resulting intervals adapt to the input, but nothing ensures that a nominal 80% interval covers 80% of new data. Conformal prediction [12, 13, 18] adds that guarantee: a held-out calibration set measures how wrong the model usually is, and the interval is widened accordingly. Under exchangeability the coverage holds in finite samples, for any model. Conformalized quantile regression (CQR) [14] combines the adaptive width of quantile regression with this guarantee. Mondrian conformal prediction [12] calibrates separately within groups, which targets coverage per group rather than on average.

Time series break exchangeability through autocorrelation and drift. Recent work addresses this with ensemble-based intervals for dependent data [16], conformal intervals for multi-horizon recurrent forecasters [17] and adaptive conformal inference (ACI) [15], which updates the miscoverage level online from observed errors. We are not aware of work that applies and compares these calibration methods for campus parking availability.

1.4 Contributions

- A reproducible forecasting pipeline for UM car park data at two horizons (1 h and 24 h), evaluated by a walk-forward simulation of November–December 2023 with strict separation of training, model selection, calibration and testing.
- Evidence that the series drifts and that its counter occasionally jumps, and a simple remedy for tree models: predicting the change from the current value instead of the level.
- A comparison of five interval methods (uncalibrated quantile XGBoost, split conformal, CQR, hour-wise Mondrian CQR and CQR with ACI) by coverage, width and interval score, overall, by hour and by academic period.
- A one-sided bound that drivers can use directly, "at least L spaces with 90% confidence", and its achieved coverage on the test period.

2 Problem Statement

Let y_t be the number of available staff car spaces at P5 in the t-th 15-minute slot (96 slots per day). At a forecast origin t we observe the history up to and including y_t, summarised in a feature vector x_t. For a horizon h we seek a point forecast and a prediction interval for y_{t+h}:

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\hat{y}_{t+h} = f(\mathbf{x}_t), \qquad \Pr\big(L_t \le y_{t+h} \le U_t\big)
\ge 1-\alpha .
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We study $h = 4$ (one hour ahead) and $h = 96$ (24 hours ahead) and use $\alpha = 0.2$, i.e. 80% intervals. A feature may use a lagged value y_{t+h-L} only if $L \geq h$, so that it is known at the origin; calendar features of the target time are known in advance.

The data are split by the time of the target, $t+h$:

Period (2023)	Role	Used to
1 Jan – 15 Sep	Training	fit model parameters
16 – 30 Sep	Validation	choose hyperparameters, early stopping, compare models
1 – 31 Oct	Calibration	compute conformal quantiles only
1 Nov – 31 Dec	Test	walk-forward simulation, reported once

In the test period the models are frozen. At each origin they see only data observed up to that time. This rolling-origin evaluation [20] reproduces real-time deployment. Point forecasts are judged by MAE and RMSE, intervals by coverage, mean width and the interval score of Gneiting and Raftery [19]. We report all-day figures and daytime figures (target time 07:00–20:00); daytime is when drivers need the forecast and when errors are large.

3 Data

3.1 Source and preprocessing

The UM Data and API platform [21] publishes the number of available spaces for each car park, vehicle type (car, motorcycle) and parking type (staff, visitor, monthly pass). We use the 2023 historical file, which covers car parks P1, P2, P3, P5 and P6 with 97,381 records per series. Records arrive about every 8 minutes in January–February and every 5 minutes from March. Our target is P5 staff car spaces; the staff series of P2, P3 and P6 enter as features. We join a calendar built from the UM academic almanacs (teaching, examination and break periods) and the 26 UM public holidays of 2023.

Preprocessing has four steps:

1. Isolated spikes are removed: a record more than 60 spaces from the median of its five-record window is set to missing (2 records for P5 staff, both on 14 July).
2. Negative counts are clipped to zero (none for P5 staff; 19 for P2 staff).
3. Records are averaged on a fixed 15-minute grid, giving 35,040 slots.

4. Gaps of up to 30 minutes are filled forward with the last observed value; longer gaps stay missing (297 slots, mainly 28–31 July). Forward filling, unlike interpolation, uses no future values.

3.2 Data quality findings

Two properties of the data shape the methods. First, the level drifts. At 03:00, when the staff car park is almost empty, the median number of available spaces is about 120 in January–May and about 150–160 from July. The night-time value also moves from day to day: its median absolute day-to-day change is 11 spaces and its 90th percentile 37.

Second, the counter jumps. In 48 cases the reported count changed by 40 or more within five minutes, mostly around 08:00 and 19:00. On 11 October, for example, the count rose from 106 to 172 at 08:20, during the morning arrival peak, and then kept falling. Such a jump cannot be cars leaving. We suspect manual counter corrections or zones being opened and closed, but the data cannot confirm either. We keep these jumps, because the quantity drivers see, and the quantity we forecast, is the displayed count.

4 Methodology

4.1 Features

Each origin t yields one row of 20 features (21 for h = 4):

Group	Features	Known at origin because
Recent values	y_t, y_{t−1}, y_{t−2}, y_{t−4}; changes over 15 min and 1 h; 1-hour mean; minimum so far today	all at or before t
Seasonal lags	same time on the previous day (h = 4 only), previous week and two weeks before, relative to the target	lag L > h
Other car parks	current staff car spaces at P2, P3 and P6	observed at t
Calendar of target	15-minute slot (index, sine, cosine), day of week, public holiday, off-day (weekend or holiday), period (teaching, exam, break)	calendar known in advance

4.2 Baselines

Three baselines need no training: persistence (y_t), seasonal naive (y_{t+h−672}, the same time last week) and a profile mean (the training-set mean for the same day type, slot and academic period).

4.3 Ridge regression

Ridge regression [10] minimises squared error plus an L2 penalty on the coefficients:

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \lVert \mathbf{y} - X\mathbf{w} \rVert^2 + \lambda \lVert \mathbf{w} \rVert^2 = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}.$$

Numeric features are standardised with training statistics; day of week, period and hour are one-hot encoded. Because a linear model cannot learn that the hour effect differs between workdays and off-days, we add their interaction as a categorical feature. λ is chosen from {0.1, 1, 10, 100, 1000} on the validation set.

4.4 Gradient-boosted trees with a change target

XGBoost [8] fits an additive model of regression trees by second-order gradient boosting [9]. Trees cannot extrapolate beyond the range of training targets, and the level of our series drifts (Section 3.2). We therefore train on the change from the current value and add it back:

$$\Delta_t = y_{t+h} - y_t, \qquad \hat{y}_{t+h} = y_t + \hat{\Delta}_t .$$

We compare this change target with the level target and depths {4, 6, 8} on the validation set. Other settings: learning rate 0.03, up to 3,000 trees with early stopping after 200 rounds without improvement in validation MAE, subsample 0.8, column sample 0.8, minimum child weight 5, L2 leaf penalty 1.

4.5 Quantile XGBoost

A single multi-quantile XGBoost model predicts the conditional quantiles q_0.1, q_0.5 and q_0.9 of the change target by minimising the pinball loss [11]

$$\rho_\tau(u) = \max\{\tau u, (\tau - 1) u\}, \qquad u = y - q .$$

The expected pinball loss is minimised at the τ-quantile, since its derivative in q is F(q) − τ. Predicted quantiles are sorted within each row to prevent crossing; [q_0.1, q_0.9] is the uncalibrated 80% interval. Depth is chosen by mean validation pinball loss.

4.6 Conformal calibration

All conformal methods use the n = 2,976 October slots, which no model has seen. With scores s_1, …, s_n, split conformal prediction [13] takes the k-th smallest score

$$\hat{q} = s_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil ,$$

which gives coverage of at least 1 − α if calibration and test points are exchangeable [12, 18]. We compare four calibrated intervals:

1. **Fixed width:** scores $|y - \hat{y}|$ of the XGBoost point forecast; interval $\hat{y} \pm \hat{q}$.
2. **CQR** [14]: scores $\max(q_{0.1} - y, y - q_{0.9})$; interval $[q_{0.1} - \hat{Q}, q_{0.9} + \hat{Q}]$.
3. **Mondrian CQR by hour** [12]: the CQR quantile is computed separately for each hour of the target (124 scores per hour), aiming at 80% coverage in every hour rather than on average.
4. **CQR with ACI** [15]: during the test period the miscoverage level is updated online, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, where $\text{err}_t = 1$ if the interval missed. For an h-step forecast, the outcome of a forecast is known only h steps later, so slot i uses errors up to slot i − h. We fix $\gamma = 0.005$ in advance.

For drivers we also build a one-sided bound: scores $q_{0.1} - y$, calibrated per hour at the 90% level, give $L = q_{0.1} - \hat{L}_{\text{hour}}$, read as "at least L spaces with 90% confidence".

4.7 Interval metrics

Besides coverage and mean width we use the interval score [19], which rewards narrow intervals and penalises misses in proportion to their size:

$$IS_\alpha = (U - L) + \frac{2}{\alpha}(L - y)\,\mathbb{1}\{y < L\} + \frac{2}{\alpha}(y - U)\,\mathbb{1}\{y > U\} .$$

5 Results

5.1 Point forecasts

XGBoost with the change target is the most accurate model at both horizons. On the validation set the change target beat the level target for XGBoost (daytime MAE 6.87 vs 7.45 at 1 h; 13.31 vs 13.68 at 24 h), and depths 8 (1 h) and 6 (24 h) were selected. For Ridge the two targets gave the same error (7.80), as expected: y_t is itself a feature, so a linear model can subtract it on its own.

Table 1. Point forecast errors in spaces. Validation = 16–30 September; test = November–December, reported once.

Horizon	Method	Val. MAE (day)	Test MAE (day)	Test RMSE (day)	Test MAE (all)
1 h	Persistence	13.19	11.22	18.37	6.50
1 h	Seasonal naive	13.20	21.08	28.05	21.66
1 h	Profile mean	31.96	27.33	31.81	27.74
1 h	Ridge	7.80	6.98	10.22	5.17
1 h	**XGBoost**	**6.87**	**6.25**	**9.76**	**3.89**
24 h	Persistence	29.10	25.86	35.29	24.17
24 h	Seasonal naive	13.20	21.08	28.05	21.66
24 h	Profile mean	31.96	27.30	31.79	27.72
24 h	Ridge	15.00	16.22	20.62	**16.94**
24 h	**XGBoost**	13.31	**16.13**	**20.34**	17.46

At 1 h, XGBoost cuts daytime test MAE by 44% relative to persistence. At 24 h it improves on seasonal naive by 23% on the test set but only matches it on the validation set (13.31 vs 13.20). Late September consisted of regular teaching weeks, where last week's profile is nearly the answer; November–December contained holidays, examinations and the Christmas recess, where it is not. Figure 1 shows that persistence lags each morning drop by an hour, while XGBoost turns with the data.

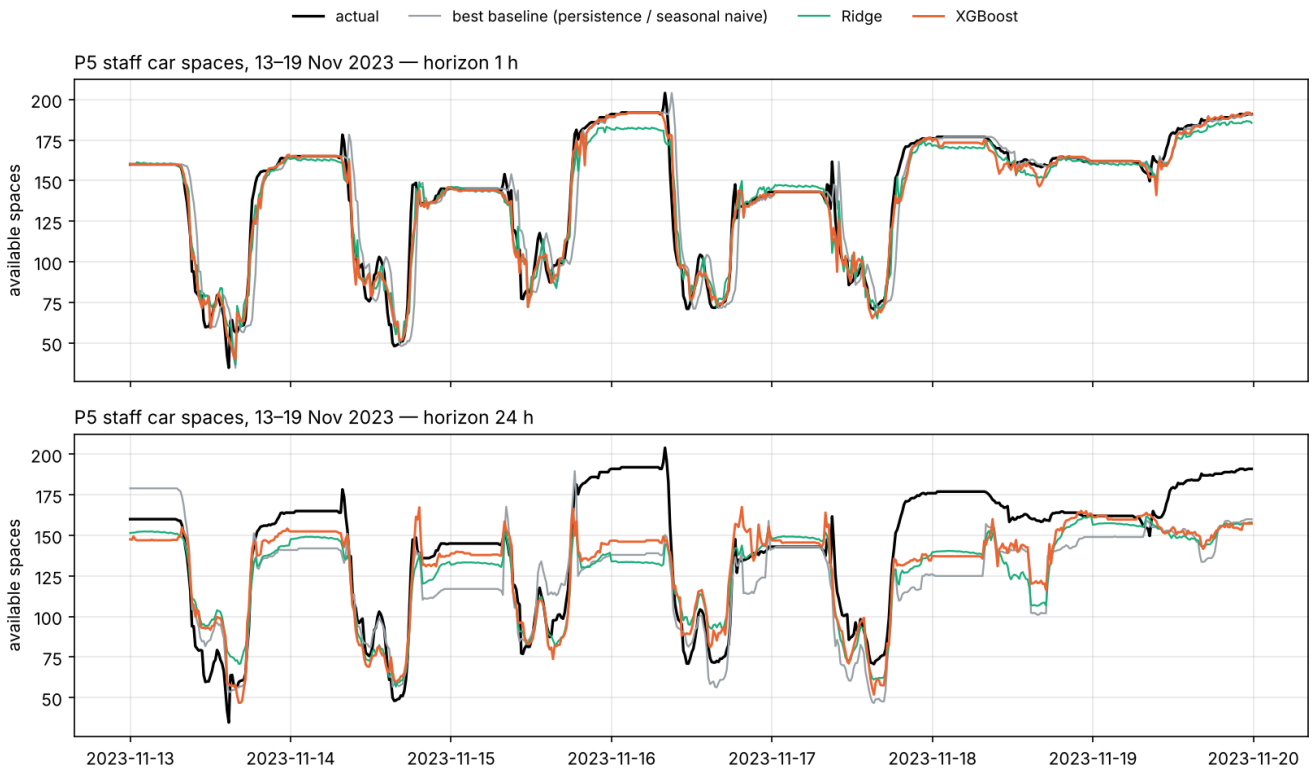


Figure 1. Actual and forecast P5 staff spaces, 13–19 November 2023. Top: 1 h ahead; bottom: 24 h ahead.

The XGBoost models rely on different information at the two horizons (gain importance, Figure 2). At 1 h the time of day (slot features, 43% of total gain) and the off-day flag (25%) dominate, because the model predicts how much the count will change in the next hour. At 24 h the off-day flag (24%) is followed by the value at the same time on the previous day (15%) and the previous week (9%).

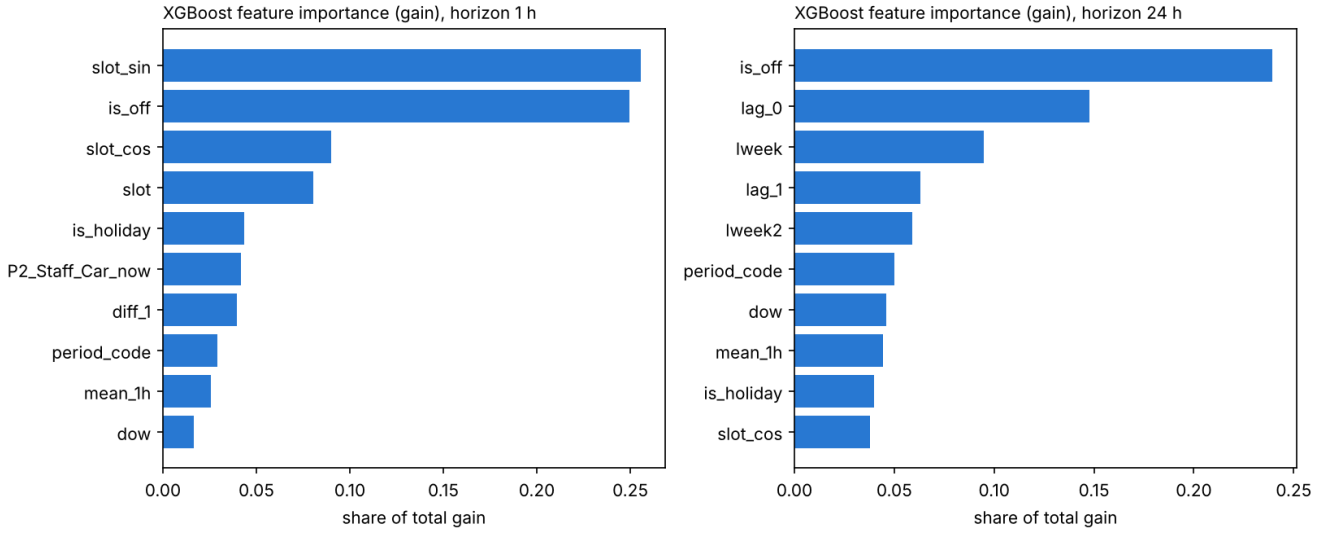


Figure 2. Share of total gain of the ten most important features.

5.2 Prediction intervals

The uncalibrated quantile intervals are too narrow, and conformal calibration repairs most of the shortfall. The October calibration gave $\hat{q}$ = 7.79 and 26.96 spaces for the fixed-width band and $\hat{Q}$ = +0.72 and +7.43 for CQR at 1 h and 24 h. The hour-wise CQR quantiles ranged from −0.08 to +13.54 (1 h) and from −3.03 to +17.47 (24 h).

Table 2. 80% intervals on the test set (November–December). Coverage target 80%; interval score: lower is better.

Horizon	Method	Coverage (all)	Coverage (day)	Width (all)	IS (all)	IS (day)
1 h	Quantile XGBoost, uncalibrated	77.3%	71.5%	9.2	19.4	31.8
1 h	Fixed width	84.0%	72.4%	15.6	28.4	37.7
1 h	CQR	85.0%	77.0%	10.7	19.6	31.4
1 h	**Mondrian CQR by hour**	86.2%	85.8%	13.1	**19.2**	**31.4**
1 h	CQR + ACI	80.5%	73.2%	9.8	19.4	31.6
24 h	Quantile XGBoost, uncalibrated	60.9%	61.1%	38.7	88.4	82.5
24 h	**Fixed width**	80.0%	83.6%	53.9	**77.8**	**72.7**
24 h	CQR	77.1%	78.9%	53.5	80.8	75.7
24 h	Mondrian CQR by hour	73.4%	78.5%	50.7	83.4	77.0

Horizon	Method	Coverage (all)	Coverage (day)	Width (all)	IS (all)	IS (day)
24 h	CQR + ACI	80.3%	80.9%	68.4	90.6	85.3

The uncalibrated intervals missed more often above than below (24 h: 23.5% above the upper bound, 15.6% below the lower bound), so the quantile model under-predicted on average. At 1 h, Mondrian CQR is the only method whose daytime coverage reaches 80%, and it has the best interval score; it over-covers somewhat. At 24 h, the fixed-width band has the best interval score and the right overall coverage. Figure 3 shows the selected interval for each horizon on a test week.

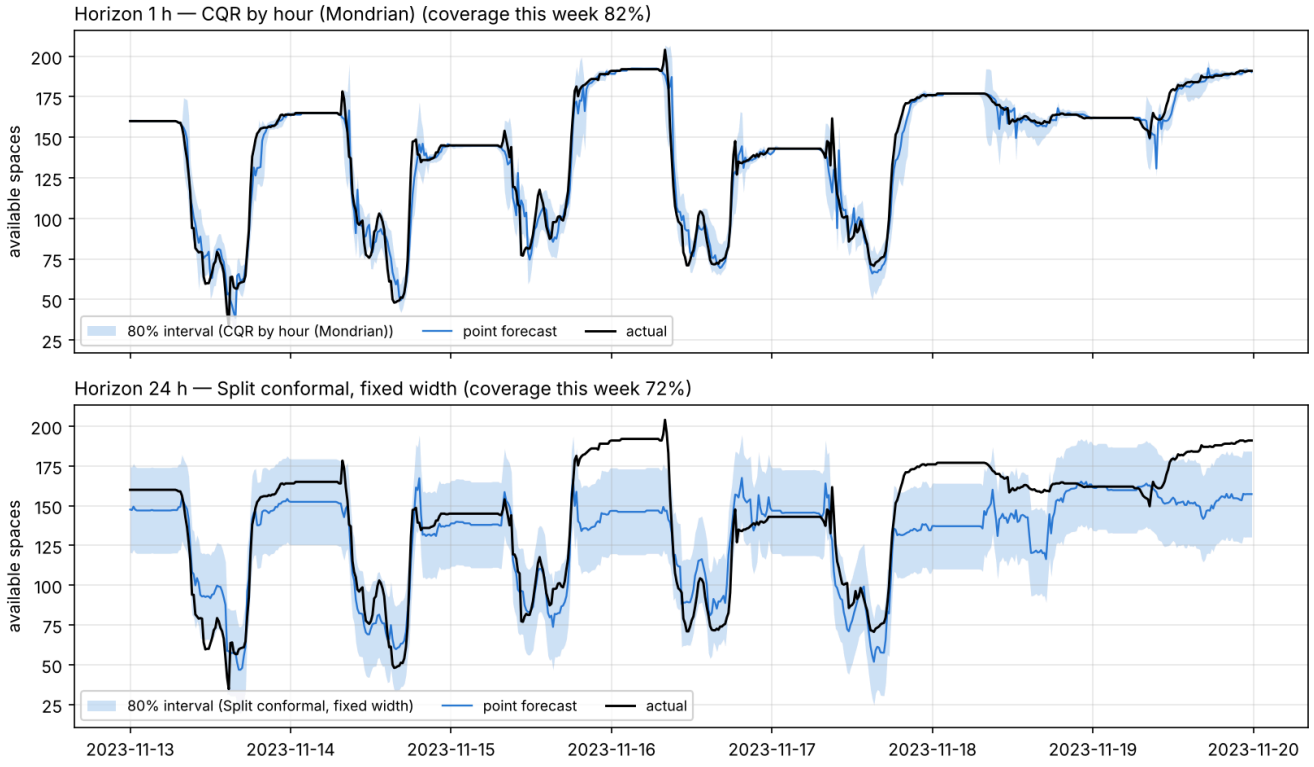


Figure 3. Selected 80% intervals, 13–19 November 2023.

5.3 Reliability, periods and the one-sided bound

Figure 4 recalibrates each method for nominal levels from 50% to 95% and plots daytime test coverage against the nominal level. At 24 h, the fixed-width band and ACI lie close to the diagonal at all levels, while CQR and Mondrian CQR under-cover at 50–70% and reach the diagonal from 80%. At 1 h, Mondrian CQR lies above the diagonal throughout, and CQR and ACI are flatter than it: they over-cover at 50% (67%) and under-cover slightly at 80%.

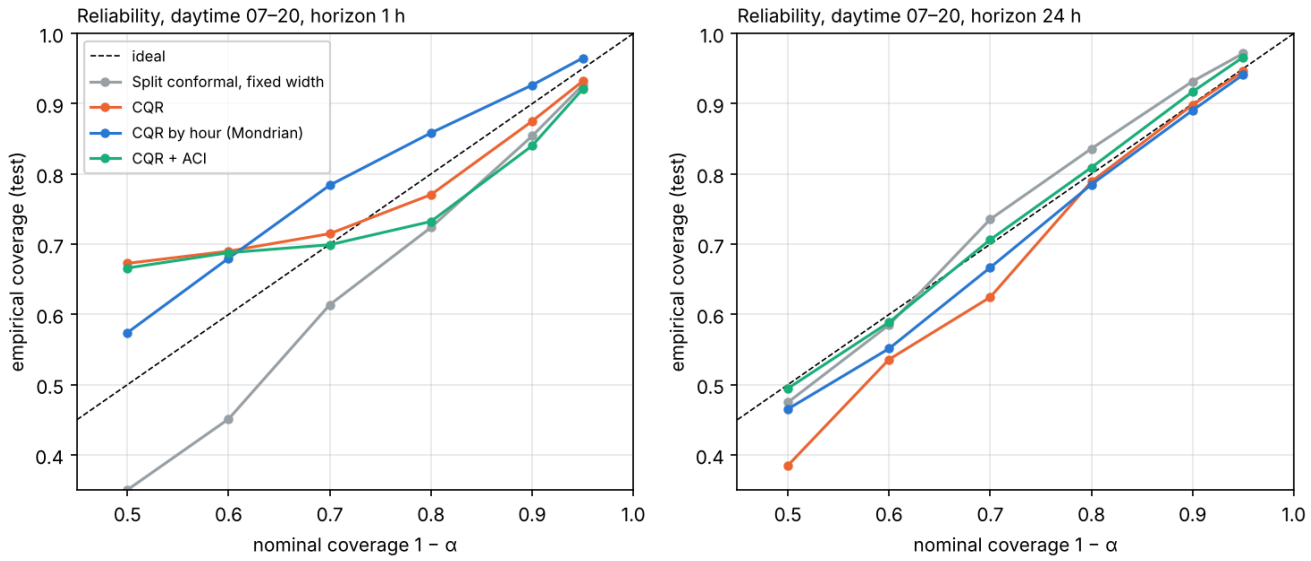


Figure 4. Nominal versus empirical daytime coverage on November–December.

Table 3. Daytime test coverage by academic period, uncalibrated versus selected method.

Horizon	Method	Teaching weekdays	Exam weekdays	Recess weekdays	Weekends and holidays
1 h	Uncalibrated	72.5%	76.0%	78.2%	67.0%
1 h	Mondrian CQR	81.0%	85.2%	90.4%	89.2%
24 h	Uncalibrated	60.5%	69.0%	83.0%	52.8%
24 h	Fixed width	80.4%	85.0%	94.9%	83.0%

Weekly daytime coverage (Figure 5) stays below 80% for the uncalibrated model in every week at 24 h and in all but the last week at 1 h, and mostly between 70% and 90% for the calibrated methods.

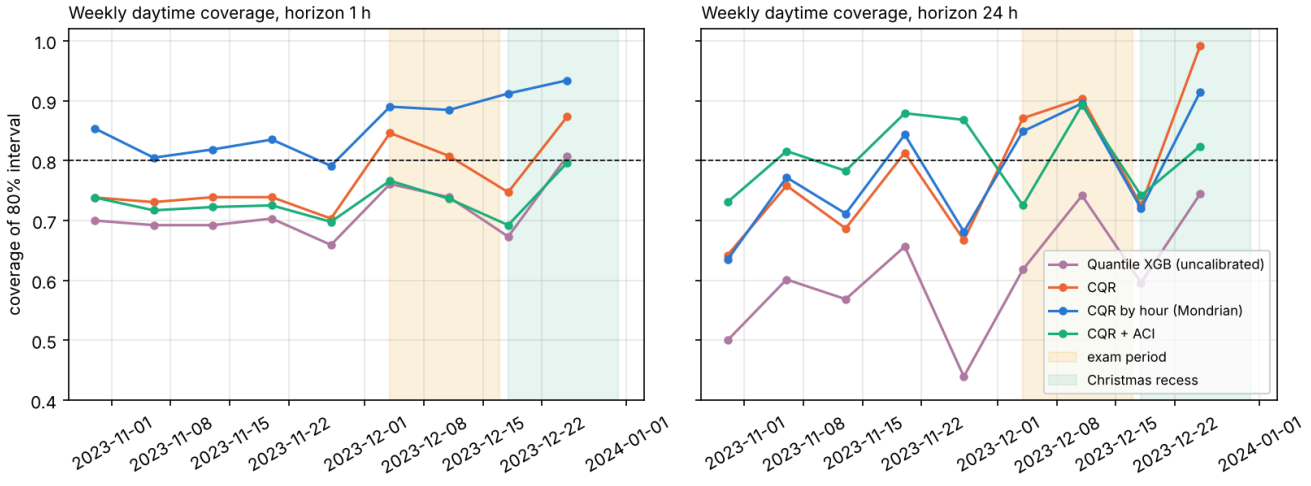


Figure 5. Weekly daytime coverage, November–December 2023; shaded: examination period and Christmas recess.

The one-sided bound "at least L spaces with 90% confidence" held for 90.6% of all test points and 92.5% of daytime points at 1 h, and for 93.1% and 90.8% at 24 h. On average the true count exceeded L by 11.3 spaces (1 h) and 25.6 spaces (24 h) in daytime. For example, at 10:00 on Monday 13 November the 24-hour forecast for Tuesday 10:00 was 85 spaces, with an 80% interval of 55–120 and a 90% lower bound of 59; the actual count was 97.

6 Discussion

6.1 Three signs of distribution shift

The level drift found in Section 3.2 shows up in three independent places. The calendar-only profile mean has almost no bias on the training set (−0.5 spaces) but under-predicts by 30.6 spaces on validation and 27.0 on test. Predicting the change rather than the level improves XGBoost at both horizons. And the uncalibrated quantile intervals miss above more often than below. Lag features and the change target let the point models follow the new level; the intervals need calibration on recent data.

6.2 Average coverage versus coverage when it matters

One global calibration constant cannot serve both night and day at 1 h. Night slots are nearly certain, so they dominate the October scores and keep the global CQR constant small (+0.72); the result covers 85% overall but 77% in daytime. Calibrating each hour separately fixes daytime coverage (85.8%) at the cost of wider intervals. It over-covers because October was harder to predict than November–December: the uncalibrated daytime coverage was 63.5% in October and 71.5% on the test set. This is a violation of exchangeability, but in the safe direction.

ACI reached the closest overall coverage (80.5% and 80.3%) because it corrects itself online. At 24 h its feedback arrives 96 steps late, so it overshoots: the miscoverage level fell close to zero in early December and the mean width grew to 68 spaces. In a sensitivity check reported only for understanding, not for selection, $\gamma = 0.001$ gave 81.5% coverage with a width of 58.1, while $\gamma = 0.01$ widened the band to 82.4.

6.3 Why a fixed band wins at 24 h

At 24 h most uncertainty concerns the day's overall level, not the time of day. The night-time level alone moves by a median of 11 spaces from one day to the next, and the 24-hour quantile intervals were wider at night (about 41) than in the afternoon (about 32). An interval that adapts to the hour therefore adds little, and the simpler fixed-width band scores best.

6.4 Failure cases

Public holidays are the clearest failure. On 20 December (Macau SAR Establishment Day) 173 spaces were free at 10:00; the 24-hour interval was 96–143 and the 1-hour interval 138–171, so both missed. Holidays are rare in training, and both the models and the calibration underrepresent them. The first days of the Christmas recess are a second case: last week's pattern was an examination week, so the seasonal naive baseline failed (daytime MAE 34.2 on recess weekdays) while the calendar-aware models did not.

7 Conclusion

Available staff spaces at UM car park P5 can be forecast with a daytime error of about 6 spaces one hour ahead and 16 spaces one day ahead, and the forecasts can carry intervals whose coverage is close to the stated level.

Key takeaways.

- The data themselves are uncertain: the level drifts by about 30 spaces over the year and the counter jumps. Predicting the change from the current value makes tree models robust to the drift.
- Quantile regression gives intervals that follow the daily rhythm but are overconfident on new data (77% and 61% coverage for a nominal 80%). Conformal calibration on held-out data is a cheap, model-agnostic fix.
- The best calibration depends on where the uncertainty comes from. Hour-wise calibration suits the 1-hour horizon, where uncertainty depends on the time of day; a fixed band suits the 24-hour horizon, where it depends on the day's level.
- A one-sided bound ("at least L spaces with 90% confidence") is the form most useful to drivers, and it held for 90.6–93.1% of test points.

Limitations. We use one car park, one year and no external data such as weather or campus events. Conformal guarantees assume exchangeability, which time series violate; our test coverage is measured, not guaranteed. The validation window is only 15 days, the hyperparameter grids are small, and the cause of the counter jumps is unknown.

Future work. Add weather, event and timetable data; forecast all car parks jointly so that drivers can be pointed to the one most likely to have space; recalibrate monthly or with rolling windows; test conformal methods built for dependent data [16, 17]; and serve live forecasts from the UM API.

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References 1–7, 16, 17 and 22 were checked against publisher or indexing pages on 1 November 2026; the group should re-check all entries before submission.

Appendix A. Use of generative AI

We used Claude (Anthropic) as a coding and writing assistant. It drafted the Python scripts for all seven steps, ran them, produced the figures and tables, wrote two explanatory documents for the group (methods and project log) and drafted this report. The group chose the topic and data, decided the evaluation design with the instructor's advice, reviewed and re-ran the code, checked the numbers in this report against the result files, and revised the text. We take responsibility for all submitted text, code, results and references, and each member can explain the parts they worked on.

Appendix B. Reproducibility

All code, the cleaned 2023 data and the result files are in github.com/taka8rie/ecen7102-carpark. Running `01_prepare.py` to `07_evaluate.py` in order regenerates every table and figure in this report; a run from a clean clone reproduced all 23 result files exactly. Software: Python 3, pandas, NumPy, scikit-learn, XGBoost 3.x (multi-quantile regression needs ⩾ 2.0) and matplotlib. Random seeds are fixed.

Appendix C. Supplementary figures and tables

These results were added after an internal review of the draft. They are produced by `08_extra.py` and stored in `results/extra/`.

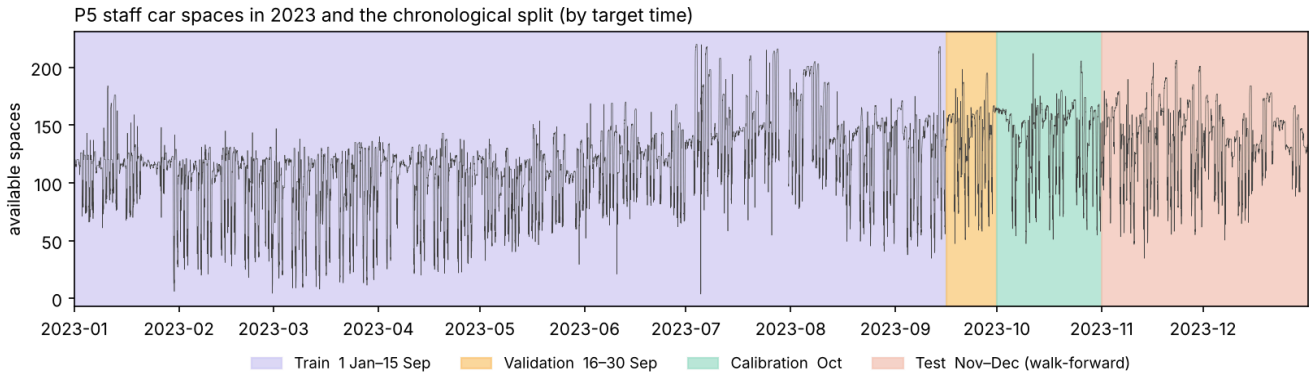


Figure C1. The 2023 series and the four periods of the chronological split (by target time). The level rises from July onward.

Table C1. Daytime MAE in spaces on the training, validation and test periods (same evaluation rows for all methods).

Horizon	Method	Train	Validation	Test
1 h	Persistence (best baseline)	11.51	13.19	11.22
1 h	Ridge	6.82	7.80	6.98
1 h	XGBoost	3.78	6.87	6.25
24 h	Seasonal naive (best baseline)	15.38	13.20	21.08
24 h	Ridge	12.79	15.00	16.22
24 h	XGBoost	8.34	13.31	16.13

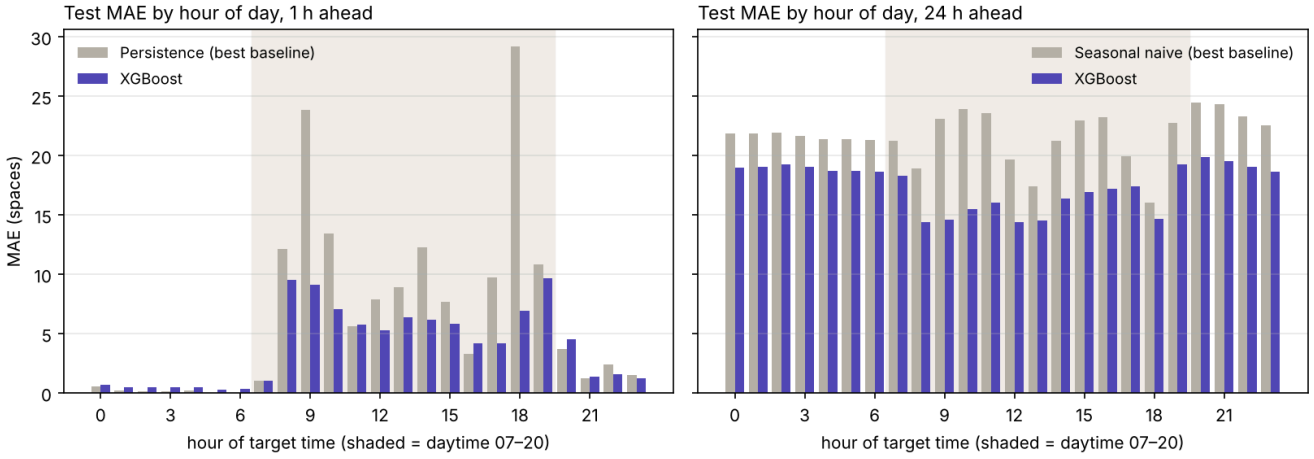


Figure C2. Test MAE by hour of the target time. At 1 h the error is concentrated in the arrival and departure peaks; at 24 h it is spread over the whole day, including the night.

Table C2. Difference in daytime test MAE (spaces) with a 95% interval from a day-block bootstrap: the 61 test days are resampled with replacement 4,000 times, so that the strong dependence within a day is preserved.

Horizon	Comparison	Difference	95% interval
1 h	XGBoost − persistence	−4.97	[−6.40, −3.47]
1 h	XGBoost − Ridge	−0.73	[−1.11, −0.28]
24 h	XGBoost − seasonal naive	−4.95	[−7.71, −2.25]
24 h	XGBoost − Ridge	−0.09	[−1.05, 0.87]